UAV Swing Payload Model

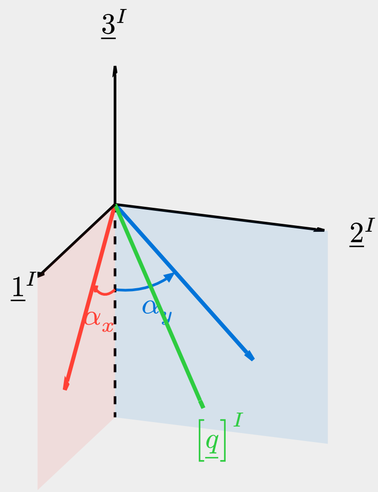
$$m_P \underline{a}_P = T\underline{q} - m_P \underline{g} \tag{1}$$

$$m_D \underline{a}_D = \underline{f} - T\underline{q} - m_D \underline{g} \tag{2}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\underline{\omega}^{BE} = \left(\underline{J}_B^B\right)^{-1}\left[\underline{n}_B - \underline{\Omega}^{BE}\underline{J}_B^B\underline{\omega}^{BE}\right] \tag{3}$$

$$\underline{s}_P = \underline{s}_D - L\underline{q} \tag{4}$$

Swing angles



$$\left[\underline{q}\right]^I = \begin{pmatrix} \cos\alpha_x & 0 & -\sin\alpha_x \\ 0 & 1 & 0 \\ \sin\alpha_x & 0 & \cos\alpha_x \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha_y & -\sin\alpha_y \\ 0 & \sin\alpha_y & \cos\alpha_y \end{pmatrix}\left[-\underline{3}^I\right]^I = \begin{pmatrix} \sin\alpha_x\cos\alpha_y \\ \sin\alpha_y \\ -\cos\alpha_x\cos\alpha_y \end{pmatrix} \approx \begin{pmatrix} \alpha_x \\ \alpha_y \\ -1 \end{pmatrix} \tag{5}$$

ArUco Marker Tracking EKF

Frames

$$\begin{aligned} [\cdot]^I &: \text{ENU} \quad (+x \text{ East}, +y \text{ North}, +z \text{ Up}) \\ [\cdot]^B &: \text{body} \quad (+x \text{ nose}, +y \text{ portside}, +z \text{ up}) \\ [\cdot]^C &: \text{camera} \quad (+x \text{ right}, +y \text{ up}, +z \text{ optical axis}) \end{aligned} \tag{6}$$

State

$$[\xi]^I = \begin{pmatrix} \alpha_x \\ \alpha_y \\ \dot{\alpha}_x \\ \dot{\alpha}_y \\ \psi_P \end{pmatrix} \in \mathbb{R}^5 \tag{7}$$

$$s_x = \sin\alpha_x, \quad c_x = \cos\alpha_x, \quad s_y = \sin\alpha_y, \quad c_y = \cos\alpha_y \tag{8}$$

α_x, α_y: tether swing angles, tipping toward $+x^I$ (East) and $+y^I$ (North) [rad]
$\dot{\alpha}_x, \dot{\alpha}_y$: swing rates [rad/s]
ψ_P: payload yaw: marker-row direction [rad]

Input

$$[T]^{IB} = \begin{pmatrix} c_\theta c_\psi & s_\varphi s_\theta c_\psi - c_\varphi s_\psi & c_\varphi s_\theta c_\psi + s_\varphi s_\psi \\ c_\theta s_\psi & s_\varphi s_\theta s_\psi + c_\varphi c_\psi & c_\varphi s_\theta s_\psi - s_\varphi c_\psi \\ -s_\theta & s_\varphi c_\theta & c_\varphi c_\theta \end{pmatrix} \tag{9}$$

Euler angles: φ: roll θ: pitch ψ: yaw

$$[\underline{a}_D]^I = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = [T]^{IB} [\underline{a}_D]^B \tag{10}$$

$[\underline{a}_D]^I$: acceleration inputs from the drone [m/ s^2]

Process model

$$\left[\dot{\xi}\right]^I = f(\xi, a) = \begin{pmatrix} \dot{\alpha}_x \\ \dot{\alpha}_y \\ -\frac{c_x a_1 + s_x a_3}{L c_y} + 2\frac{s_y}{c_y}\dot{\alpha}_x\dot{\alpha}_y \\ -\frac{c_y a_2 + s_y(c_x a_3 - s_x a_1)}{L} - s_y c_y \dot{\alpha}_x^2 \\ 0 \end{pmatrix} \approx \begin{pmatrix} \dot{\alpha}_x \\ \dot{\alpha}_y \\ -\frac{1}{L}(a_1 + \alpha_x \|g\|) \\ -\frac{1}{L}(a_2 + \alpha_y \|g\|) \\ 0 \end{pmatrix} \tag{11}$$

L: drone tether-pivot to payload CG [m]

$$F = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -\frac{\|g\|}{L} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\|g\|}{L} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \tag{12}$$

$F \in \mathbb{R}^{5\times5}$: process Jacobian (continuous-time)

Prediction

Jacobian to discrete-time conversion w/ Euler integration.

$$\hat{\xi}_{k+1}^- = \hat{\xi}_k^+ + \Delta t\, f\left(\hat{\xi}_k^+, a_s\right) \tag{13}$$

$$\Phi_k = I_5 + F\Delta t, \qquad Q = \operatorname{diag}(0, 0, q_x, q_y, q_\psi)\Delta t \tag{14}$$

$$P_{k+1}^- = \Phi_k P_k^+ \Phi_k^\top + Q \tag{15}$$

Δt: time step [s]
$\hat{\xi}_k^-, \hat{\xi}_k^+$: estimate before & after the update at step k
$P_k^-, P_k^+ \in \mathbb{R}^{5\times5}$: covariance before & after
Φ_k: discrete time Jacobian
q: process-noise intensity

Shelving Jacobian to discrete-time conversion w/ RK4 for now.

$$\begin{aligned} k_1 &= f\left(\hat{\xi}_k^+, a_s\right), \quad k_2 = f\left(\hat{\xi}_k^+ + \frac{\Delta t}{2} k_1, a_s\right) \\ k_3 &= f\left(\hat{\xi}_k^+ + \frac{\Delta t}{2} k_2, a_s\right), \quad k_4 = f\left(\hat{\xi}_k^+ + \Delta t\, k_3, a_s\right) \end{aligned} \tag{16}$$

$$\hat{\xi}_{k+1}^- = \hat{\xi}_k^+ + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{17}$$

$$\Phi_k = I_5 + F\Delta t + \frac{1}{2}(F\Delta t)^2, \qquad Q = \operatorname{diag}(0, 0, q, q, q_\psi)\Delta t \tag{18}$$

$$P_{k+1}^{-} = \Phi_k P_k^{+} \Phi_k^{\top} + Q \tag{19}$$

Δt: time step [s]
$\hat{\xi}_k^-, \hat{\xi}_k^+$: estimate before & after the update at step k
$P_k^-, P_k^+ \in \mathbb{R}^{5\times5}$: covariance before & after
Φ_k: discrete time Jacobian
q: process-noise intensity

Measurement

Pre-processing logic

For every $j \in V_k$, `solvePnP` returns the marker pose in the camera frame,

$$\left[\underline{t}_j\right]^C, \quad [T]^{CM_j} \qquad j \in V_k \tag{20}$$

V_k: marker IDs detected in frame k
$\left[\underline{t}_j\right]^C$: marker j center in the camera frame [m]
$[T]^{CM_j}$: transformation to the camera frame from the marker local frame M_j

The first column of $[T]^{CM_j}$ is the marker x-axis, which runs along the marker row,

$$\left[\underline{m}_j\right]^C = [T]^{CM_j} \left[\underline{1}^{M_j}\right]^{M_j} \tag{21}$$

Each marker gives its own estimate of the board center; average over the markers actually detected in the frame,

$$\left[\underline{p}_{\text{ctr}}\right]^C = \frac{1}{m_k} \sum_{j \in V_k} \left(\left[\underline{t}_j\right]^C + o_j \left[\underline{m}_j\right]^C \right) \tag{22}$$

$\left[\underline{p}_{\text{ctr}}\right]$: payload center in the camera frame
m_k: # of markers in the frame
o_j: marker offset from board center along the row, +left [m]

Shift the origin from the camera optical center to the tether pivot,

$$\left[\underline{p}\right]^C = \left[\underline{p}_{\text{ctr}}\right]^C + [T]^{CB}\left(\left[\underline{t}_{BC}\right]^B - \left[\underline{\ell}\right]^B\right) \tag{23}$$

$\left[\underline{p}\right]^C$: payload center after the camera offset is corrected, camera frame [m]
$\left[\underline{t}_{BC}\right]^B$: camera optical center in the body frame [m]
$\left[\underline{\ell}\right]^B$: drone tether pivot in the body frame [m]

Average the payload yaw and transform into the inertial frame,

$$\begin{gathered} [\underline{m}]^I = [T]^{IB}[T]^{BC}\left(\frac{1}{m_k}\sum_{j \in V_k} \left[\underline{m}_j\right]^C\right) \\ z_{\psi_P} = \operatorname{atan2}\left(m_2^I, m_1^I\right) \end{gathered} \tag{24}$$

Measurment model

After some pre-processing from pixels to camera frame coordinates, we measure,

$$z_k = \begin{pmatrix} \frac{p_x^C}{\|\underline{p}\|} \\ \frac{p_y^C}{\|\underline{p}\|} \\ z_{\psi_P} \end{pmatrix} + \varepsilon_k \tag{25}$$

$\underline{p}$: payload center position
z_{ψ_P}: measured payload yaw
ε_k: measurement noise

Prediction of measurement

Measurement predicted from swing angles,

$$h(\xi) = \begin{pmatrix} q_x^C \\ q_y^C \\ \psi_P \end{pmatrix} \qquad \left[\underline{q}\right]^C = [T]^{CB}[T]^{BI} \begin{pmatrix} \alpha_x \\ \alpha_y \\ -1 \end{pmatrix} \tag{26}$$

$$H = \frac{\partial h}{\partial \xi} = \begin{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} [T]^{CB}[T]^{BI} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{pmatrix} \tag{27}$$

Update

Innovation

$$y_k = z_k - h_k\left(\hat{\xi}_k^-\right) \tag{28}$$

$$R_k = \operatorname{diag}\left(\sigma_x^2, \sigma_y^2, \sigma_\psi^2\right) \in \mathbb{R}^{3\times 3} \tag{29}$$

$$S_k = H_k P_k^- H_k^\top + R_k \in \mathbb{R}^{3\times 3} \tag{30}$$

y_k: innovation
R_k: measurement noise
S_K: innovation covariance

Kalman gain

$$K_k = P_k^- H_k^\top S_k^{-1} \in \mathbb{R}^{5\times 3} \tag{31}$$

$K_k \in \mathbb{R}^{5\times 3}$: Kalman gain

$$\hat{\xi}_k^+ = \hat{\xi}_k^- + K_k y_k, \tag{32}$$

$$P_k^+ = (I_5 - K_k H_k) P_k^- \tag{33}$$

If no markers detected:

$$\hat{\xi}_k^+ = \hat{\xi}_k^-, \quad P_k^+ = P_k^- \tag{34}$$

Control

State-Space

$$\dot{x} = Ax + Bu \tag{35}$$

where,

$$x = \begin{pmatrix} s_1^I \\ s_2^I \\ s_3^I \\ v_1^I \\ v_2^I \\ v_3^I \\ \alpha_x^I \\ \alpha_y^I \\ \dot{\alpha}_x^I \\ \dot{\alpha}_y^I \end{pmatrix}, \quad u = \begin{pmatrix} a_1^I \\ a_2^I \\ a_3^I \end{pmatrix} \tag{36}$$

$$\text{drone states}\begin{cases}\dot{s}_1^I = v_1^I \\ \dot{s}_2^I = v_2^I \\ \dot{s}_3^I = v_3^I \\ \dot{v}_1^I = a_1^I \\ \dot{v}_2^I = a_2^I \\ \dot{v}_3^I = 0\end{cases} \tag{37}$$

$$\text{payload states}\begin{cases}\dot{\alpha}_x^I = \dot{\alpha}_x^I \\ \dot{\alpha}_y^I = \dot{\alpha}_y^I \\ \ddot{\alpha}_x^I = -\frac{g}{L}\dot{\alpha}_x^I - \frac{1}{L}a_1^I \\ \ddot{\alpha}_y^I = -\frac{g}{L}\dot{\alpha}_y^I - \frac{1}{L}a_2^I\end{cases} \tag{38}$$

$$\Rightarrow A, B \tag{39}$$

Equilibrium

$$x^* = \begin{pmatrix} [s_{PL}(t)]^I + \begin{pmatrix} 0 \\ 0 \\ L \end{pmatrix} \\ [v_{PL}]^I \\ 0 \end{pmatrix} \tag{40}$$

LQI Controller

Control Law

$$u = -(K_x \;\; K_I)\begin{pmatrix} e \\ \int e \end{pmatrix} \tag{41}$$

$$e = x - x^*$$

$$y_{\text{error}} = Ce \tag{42}$$

$$\int e[k+1] = \int e[k] + y_{\text{error}}\Delta t \tag{43}$$

Output

$$y = Cx \tag{44}$$

$$\begin{aligned} y &= [s_{PL}]^I \\ &= \begin{pmatrix} s_1^I - L\alpha_x^I \\ s_2^I - L\alpha_y^I \\ s_3^I - L \end{pmatrix} \end{aligned} \tag{45}$$

$$\Rightarrow C \tag{46}$$

Gain matrix

$$\overline{K} = (K_x \;\; K_I) \tag{47}$$

Augmented State-space

$$\dot{z} = \overline{A}z + \overline{B}u \tag{48}$$

$$z = \begin{pmatrix} x \\ \int [s_{PL}]^I \end{pmatrix} \tag{49}$$

$$\Rightarrow \overline{A}, \overline{B} \tag{50}$$

Cost minimization

Solve for the optimal $\overline{K}$,

$$J = \int_0^\infty \left(\begin{pmatrix} e \\ \int e \end{pmatrix}^\top \overline{Q} \begin{pmatrix} e \\ \int e \end{pmatrix} + u^\top R u \right) \mathrm{d}t \tag{51}$$

State weighting matrix,

$$\overline{Q} = \begin{pmatrix} C^\top W C & 0_{10\times3} \\ 0_{3\times10} & W_{\text{int}} \end{pmatrix} \in \mathbb{R}^{13\times13} \tag{52}$$

$$W = \begin{pmatrix} w_x & 0 & 0 \\ 0 & w_y & 0 \\ 0 & 0 & w_z \end{pmatrix} \tag{53}$$

$$W_{\text{int}} = \begin{pmatrix} w_{\text{int},x} & 0 & 0 \\ 0 & w_{\text{int},y} & 0 \\ 0 & 0 & w_{\text{int},z} \end{pmatrix} \tag{54}$$

Control weighting matrix,

$$R = (\text{tuning const}) \times \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{55}$$

$$\overline{A}^\top P + P\overline{A} - P\overline{B}R^{-1}\overline{B}^\top P + \overline{Q} = 0 \Rightarrow P \tag{56}$$

$$\overline{K} = R^{-1}\overline{B}^\top P \tag{57}$$